Week 2 Progress

Recalibrated science; shared next steps

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What this update is for

- Share how the **17 July** meeting changed the scientific target
- Show what public data layers are now in place
- Keep weather definitions **open for team judgment** — especially Hogan
- Be clear what still depends on Roro's aggregates — and on working together

Tone: readiness is not a solo finish line. Listening and give-and-take are part of the work.

The story in one sentence

The meeting told us what the data cannot be.

The week prepared what we can analyse — without closing definitions the team still owns.

Heard in the meeting

- No admission reasons in the general HA file
- Near-term outcome: stroke aggregates
- Explore many methods, not one hero model

Built carefully

- Real pollution and flu layers
- A labelled multi-method panel
- Plumbing tested only on synthetic stroke

What the lab meeting changed

Before	Meeting	Now
AMI + IS + HS principal-dx panel	Reasons for admission unavailable	Stroke aggregates; AMI out of scope from that file
One primary heat model	“Try ∼10 methods”	Labelled pathway panel
Heatwave metrics of interest	Align with Chao Ren / Wang EHWE	Spells, 2D3N, multi-def months — still discussable
Parallel to historical Goggins	Lab daily mortality HW work exists	Complementary estimand, not a duplicate

Insight: when outcomes are aggregate and heatwave definitions are contested, specification diversity is the design — and definitions should be shared, not frozen alone.

Recalibrated question

Among Hong Kong residents in the eligible aggregate population, how do **monthly** thermal exposures associate with **stroke admission aggregates** (2013–2023), and how stable are those associations across a pre-labelled panel of specifications?

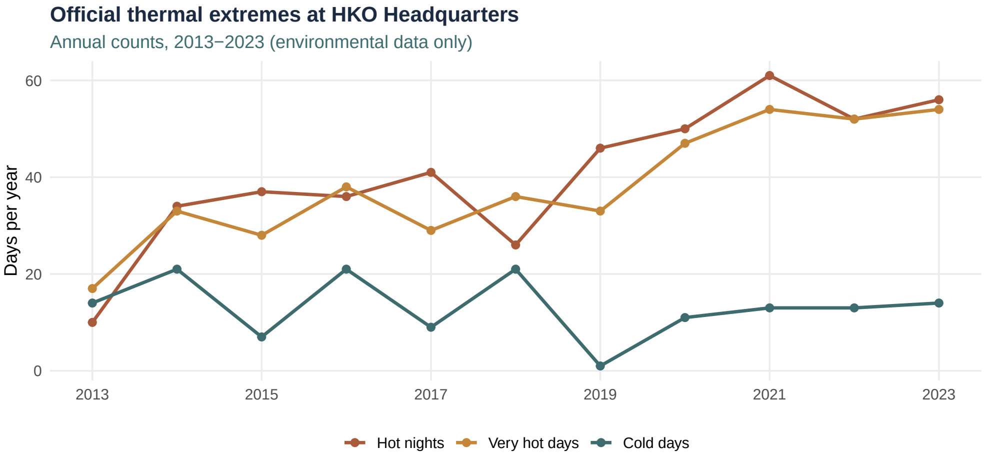
This estimand

- Monthly hospital burden
- Stroke aggregates
- Panel of labelled pathways

Not this estimand

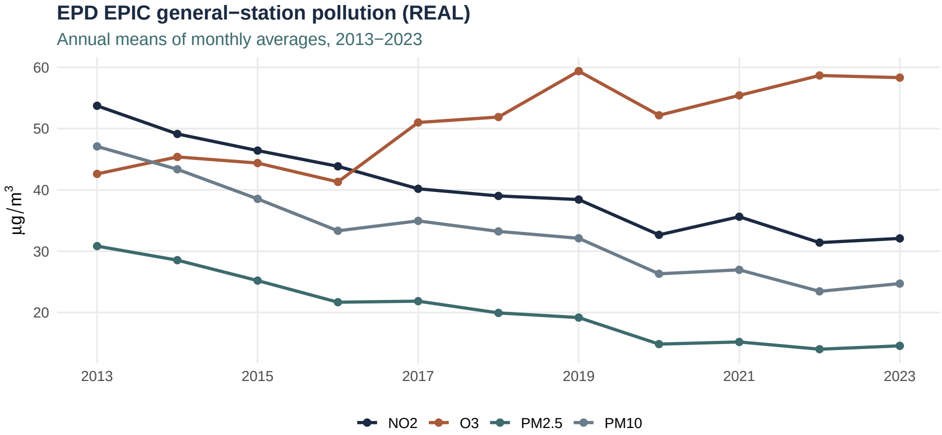
- Daily DLNM triggering (Goggins)
- Excess-death heatwave counts
- AMI from files without reasons

Climate layer: shared ground truth



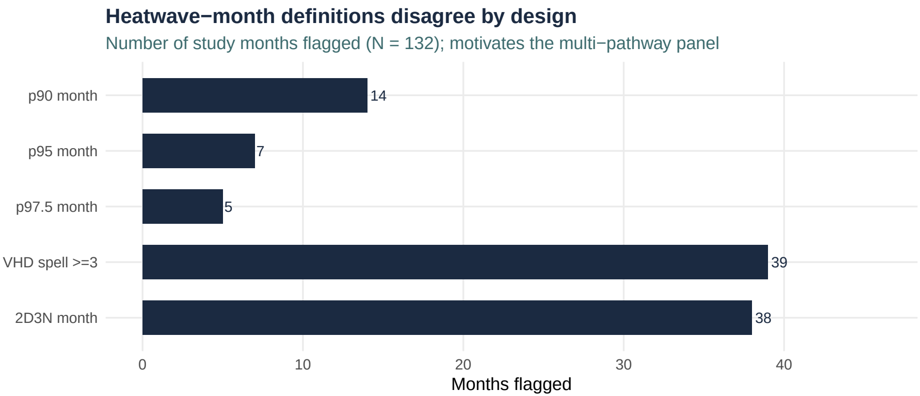
Hot nights and very hot days rose after 2018; cold days persist. Peak hot-night year in-window: **2021 (61)**. These are environmental descriptives only — not admission findings.

Pollution is now real (and staged carefully)



EPD EPIC general-station means, 2013–2023. NO_2 and PM declined; O_3 rose. Modelling rule: stage pollutants; enter O_3 last because of heat–ozone collinearity.

Why heatwave definitions should stay plural



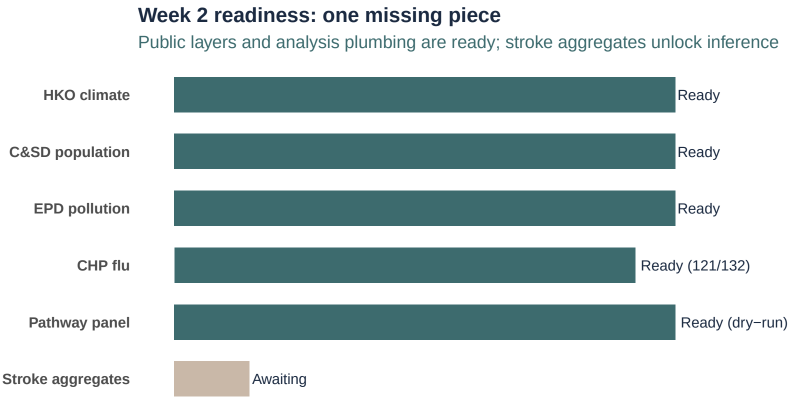
Different definitions flag different months — by design. Guo/Gasparrini, Wang/Ren, and the lab's multi-definition mortality work all argue against one universal heatwave rule. **Hogan's hot-month idea** (heatwave counts → upper-tail months) belongs in this conversation, not outside it.

A labelled panel — not seventeen discoveries

Family	Examples
Continuous temperature / lag	Tmean; Tmax/Tmin; lag-1
Extremes & heatwaves	Official counts; spells; 2D3N; HW-month indicators
Cold	Cold-day / cold-month counterparts
Structure	Age; sex (if grain allows)
Sensitivities	COVID/holidays; pollution; humidity; flu; pre-COVID window
Optional	IS vs HS only if subtype exists

Reporting rule: one panel. Gate 3 freezes a headline pair *with the team* after real descriptives (working proposal: Tmax/Tmin + official extreme-day counts).

What is ready — and what still needs people



Public layers and analysis plumbing can advance now. **Stroke aggregates** still need Roro's transfer path. **Weather / hot-month definitions** still need Hogan's judgment before anything is treated as settled.

Honesty about the dry-run

What the dry-run showed

- The multi-method panel runs end-to-end
- Missing subtype correctly disables that pathway
- Flu uses complete-case months where coverage exists

What it does not mean

- Coefficients are **not** findings
- Synthetic stroke is only a rehearsal
- Definitions are not frozen without the team

Parallel to Week 1: real environmental descriptives can be shared; association claims wait for real outcomes — and for shared specification choices.

Literature that keeps us honest (selected)

- **Goggins 2012/2013** — cold-dominant patterns; subtype and sex matter
- **Guo 2024** — official hot-night binary often weak; intensity / structure matter
- **Wang/Ren 2019** — spells and combined day–night windows
- **Guo/Gasparrini 2017** — percentile × duration grids; no single heatwave definition
- **Yang/Chong 2025** — cold and influenza with stroke admissions
- **Guo 2025** — pollution can amplify temperature–hospitalisation associations

How we want to work from here

Bob

- Keep the analysis panel transparent and labelled
- Listen before freezing headline specs
- Write meeting and team decisions into the repo

Team

- **Hogan:** weather / heatwave / hot-month definitions; environmental interpretation
- **Roro:** stroke aggregates, dictionary, transfer rules
- **Dr Bishai:** governance, Gate 3 taste, teamwork norms

> *If you want to go fast, go alone; if you want to go far, go together.*
> — guidance we are taking seriously.

Still open

Open item	Working default	Owner
Stroke aggregates + dictionary	Minimum: month and counts	Roro / Bob
Grain (territory vs age×sex)	Prefer age×sex if present	On inspection
Hot-month / heatwave ops	Include Hogan's count→percentile idea as a named option	Hogan / Bob
Headline pair (Gate 3)	Proposal: Tmax/Tmin + official extremes	Team
Governance / use path	Follow PI determination	Dr Bishai

Closing

The analysis can be prepared carefully.
The judgments should still be shared.

- Public layers are real and usable
- The multi-method panel is ready to receive aggregates
- Weather definitions remain a team conversation
- No synthetic coefficients as findings

Repo: `https://github.com/bobshenruililin/Laidlaw-Heat-Project`

Appendix

Numbers and pathway index

A1 — Environmental numbers (descriptive)

- Study months: $N = 132$ (Jan 2013–Dec 2023)
- Mean temperature $\approx 24.0^{\circ}$C
- Lag-1 Tmax complete for all 132 months
- Example HW-month flags: p90 $= 14$, p95 $= 7$, p97.5 $= 5$; 2D3N months $= 38$
- Flu coverage: 121/132 months (early-2013 gap)
- Pollution: EPD general stations (real series)

A2 — Pathway families (short index)

Continuous T / lag; official extremes; Ren/Wang spells & 2D3N; multi-def HW-month; cold-side; age/sex structure; COVID/holiday, pollution, humidity, flu sensitivities; temperature variability; pre-COVID window; subtype only if present.

Full machine registry: `analysis_plan/pathway_registry.yml`.